Pooja Sethi

Research Engineer & Technical Lead

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Summary

Research Engineer and Technical Lead specializing in foundation model post-training, multimodal evaluation, and large-scale generative AI systems. Proven track record building evaluation infrastructure, curating high-quality training data, and shipping post-trained models to billions of users at Meta. Experience spans multimodal architectures (vision-language, speech-text), LLM post-training and distillation, and rigorous benchmark design, with deep expertise bridging research and production.

Experience

Oct 2022–Present — **Senior Research Engineer & Technical Lead, Machine Translation**, *Meta AI Content Understanding*, Seattle, WA

- Technical lead for a 10+ person team developing Meta's next-generation translation systems, replacing legacy NMT with LLM-based generative models serving O(1B+) daily requests across Instagram and Facebook.
- Spearheaded research-to-production for FAIR's Seamless multimodal speech-to-speech system; led post-training and STT evaluation across 16 languages, shipping the voice dubbing feature in Instagram Reels (featured at Meta Connect 2024).
- Shipped an 8B model to 100% of Instagram Reels caption traffic, meeting strict latency/cost constraints while driving statistically significant gains in user dwell time and reshares.
- Developed competitive low-resource generative models (Indic/African languages) outperforming Gemini on Facebook posts via targeted continuous pre-training and post-training.
- Scaled Direct Preference Optimization (DPO) recipe utilizing Machine Translation Post-Editing (MTPE) data flywheel, correcting model biases and outperforming teacher demonstrations.
- Built comprehensive evaluation infrastructure spanning 100+ languages; automated quality-cost evaluation across 1P/2P/3P models across translation metrics (BLEU, MetricX, chrF); designed human evaluations and production A/B testing experiments.

Oct 2021–July 2022 — **Senior Software Engineer 2, Machine Learning**, *Impira (Acquired by Figma)*, San Francisco, CA

- First NLP engineer at Series B startup (30 people); reported to CTO.
- Developed and shipped vision-language models (LayoutLM) combining text and visual features, enabling tasks like unsupervised document clustering and information extraction.
- Designed benchmark datasets and evaluation metrics for document similarity and clustering quality.
- Research demo and evaluation methodology at: github.com/poojasethi/visual-doc-clustering.

Oct 2017–Oct 2021 — **Senior Software Engineer, Machine Learning**, *Facebook Reality Labs*, Redmond, WA

- Early engineer on Natural Language Understanding team for Facebook Assistant (Portal, Oculus, Ray-Ban Meta devices).
- Built AutoNLU, an active learning and evaluation platform for semantic parsing. Developed error detection algorithms identifying model failure modes and root-cause analysis tooling.
- Designed and implemented evaluation pipelines for intent classification and slot filling models, including adversarial test sets and cross-domain generalization benchmarks.
- Conducted 100+ technical interviews. Bootstrapped and grew internal mentorship program to 70 participants.

Research Experience

Sept 2016–Mar 2017 **Research Assistant**, *UW Natural Language Processing (NLP) Group*, Seattle, WA
- Proposed new NLP task: conversational question generation (CQG). Work received Facebook ParlAI research award ($19,800).
- Developed RNN-based question generation models with evaluation framework including automated metrics (BLEU, ROUGE) and human evaluation protocols.
- Advised by Professor Yejin Choi.

Apr 2014–June 2017 **Research Assistant**, *UW ICT for Development (ICTD) Lab*, Seattle, WA
- Developed Respeak, an Android crowdsourcing application for low-income blind users in India.
- Created evaluation benchmarks demonstrating 6X improvement in Hindi and 3X improvement in Indian-accented English ASR accuracy.
- Designed user study methodology and quality metrics for voice-based task completion. Published evaluation results at CHI (Best Paper Honorable Mention).
- Advised by Professor Richard Anderson and Aditya Vashistha.

Mar 2015–June 2015 **Visiting Undergraduate Researcher**, *Allen Institute for AI (AI2)*, Seattle, WA
- Explored adversarial perturbations of science questions to systematically increase or decrease difficulty for Q&A models.
- Developed evaluation framework for measuring model robustness and identifying failure modes.
- Advised by Been Kim (now at Google Brain).

Education

2021–2025 **M.S. in Computer Science, AI Concentration**, *Stanford University*, GPA: 3.94/4.0
Honors Cooperative Program
Select Coursework: Deep Reinforcement Learning (CS 224R), Natural Language Processing (CS 224N), Trustworthy Large Language Models (CS 329T), Spoken Language Processing (CS 224S), Deep Learning for Computer Vision (CS 231n), Deep Multi-task and Meta-learning (CS 330), Convex Optimization (EE 364a), Decision Making Under Uncertainty (CS 238), Machine Learning Under Distribution Shifts (CS 329D)

2013–2017 **B.S. in Computer Engineering**, *University of Washington, Seattle*, GPA: 3.85/4.0
Magna Cum Laude with Department Honors
Awards: Best Undergraduate Honors Thesis Award; Outstanding Computer Engineering Senior Award (top two graduating seniors).

Technical Skills & Expertise

ML/AI LLM post-training & evaluation, RLHF, multimodal architectures (vision-language, speech-text), model distillation, transformer architectures (Llama, LayoutLM, encoder-decoder)

Data Curation Large-scale dataset creation and curation, multilingual data collection, organic and synthetic data generation, data quality assessment, Hive and ETL pipelines

Evaluation Benchmark design and curation, automated evaluation pipelines, human evaluation protocols, A/B testing, quality metrics (BLEU, MetricX, chrF), adversarial testing

Engineering PyTorch, Python, HuggingFace trainers, distributed training, research-to-production workflows

Selected Publications

Preprint **AutoNLU: Detecting, root-causing, and fixing NLU model errors.** Pooja Sethi, Denis Savenkov, et al.

CHI 2018 **BSpeak: An Accessible Voice-based Crowdsourcing Marketplace for Low-Income Blind People.** Aditya Vashistha, Pooja Sethi, and Richard Anderson.

CHI 2017 **Respeak: A Voice-based, Crowd-powered Speech Transcription System.** Aditya Vashistha, Pooja Sethi, and Richard Anderson. *(Best Paper Honorable Mention - top 5%)*

Selected Awards & Honors

2017 Facebook ParlAI Research Grant ($19,800 for conversational question generation research)

2017 Outstanding Computer Engineering Senior Award (top graduating senior)

2017 Best Undergraduate Honors Thesis Award

2013 UW CSE Freshman Direct Admit (<5% acceptance rate)

2013 Admitted to MIT class of 2017, Early Action (declined)